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Subject: Cryptography and Network Security

Set-3
Assignment - 2

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- Q1. Describe the full process of Triple DES encryption using two keys:
 $K_1 = 101010$ $K_2 = 010101$ on data Block $D = 110011$.

Triple DES is an enhanced version of DES that applies DES algorithm 3 times to improve security. When key one used encrypt - decrypt - encrypt is followed.

Given

Block $D = 110011$

Key 1 = 101010

Key 2 = 010101

92
100
@

Encryption Steps:

Step 1: encrypt K_1

$$C_1 = E(K_1, D) = E(101010, 110011)$$

Step 2: Decrypt K_2

$$C_2 = D(K_2, C_1) = D(010101, C_1)$$

Step 3: Encryption K_1

$$E = (K_1, K_2) = E(101010, C_2)$$

$$E = (K_1, D(K_2, E(K_1, D)))$$

↑

Mathematical representation

Decrypt process

Step 1: Decrypt with K_1

$$C_2 = D(K_1, \text{Cipher text})$$

Step 2: Encrypt with K_2

$$C_1 = E(K_2, C_2)$$

Step 3: Decrypt with K_1

$$\text{Plaintext} = D(K_1, C_1)$$

Formula

$$\text{Plaintext} = D(K_1, E(K_2, D(K_1, \text{Cipher text})))$$

Advantage

- * Provide about 112-bit effective security.
- much storage than single DES
- Widely used in banking and payment systems.

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Security and Network

is meet in the middle attack? illustrate how it applies to
using Double DES.

Meet in the middle Attack is a cryptographic attack that reduces
effort required to break Double DES. Instead of trying every
possible pair of key directly, the attacker computes encryption from
one side and decryption from other until they meet.

Double DES

Encryption:

$$C = E(K_2, E(K_1, P))$$

where

$P \rightarrow$ plain text, $C =$ cipher text, K_1 and K_2 are two DES keys

Attack Steps

Suppose an attacker knows plain text and cipher text

1: Encrypt the plain text using every possible $K_1 \rightarrow E(K_1, P)$

2: Decrypt the cipher text using every possible $K_2 \rightarrow D(K_2, C)$

Compare both

$$E(K_1, P) = D(K_2, C)$$

Then corresponding keys are possible candidates.

Example

$$P = 110011$$

$$C = 001101$$

The attacker compute $\rightarrow E(K_1, 110011)$

for all K_1 and $D(K_2, 001101)$ and K_2

Wherever both matches the attackers find the correct key pair

Complexity:

* Brute force: 2^{112} operations

* MITM attack: 2^{57} operation

Thus, Double DES is much weaker than expected because of this

3.3 RSA problem : encrypt and decrypt message $M=5$ using $p=3$, $q=11$

$$p=3, M=5, q=11, e=7$$

Step 1: Calculate n

$$n = p \times q = 3 \times 11 = 33$$

Step 2: Calculate Euler's totient

$$\phi(n) = (p-1)(q-1) \Rightarrow (3-1)(11-1) \Rightarrow (2)(10) = 20$$

Step 3: Public key

$$(e, n) \Rightarrow (7, 33)$$

ing in block cipher. Show how padding would be added to
"hello" in 8-byte blocks

the process of adding extra bytes to end of message so that
become an exact multiple of block size.

necessary because block cipher (DES, AES) encrypt fixed size.

Block Size: 8 bytes
Required bytes: $8 - 5 = 3$ bytes

padding

Since 3 bytes are missing each added byte has value 03
al:

HELLO

padding:

HELLO 03 03 03

nal form:

E	L	L	O	pad	pad	pad
45	4C	4C	4F	03	03	03

are of padding

ata fit the required block size

encryption errors

Introduce the use of elliptic - curve Cryptography for secure messaging in IoT devices

elliptic - curves Cryptography is a public-key technique based on mathematical formula of elliptic - curves. It provides strong security while using smaller key sizes than RSA. This makes it ideal for IoT.

Why ECC is Suitable for IoT

- ★ Smaller key sizes reduce memory usage
- ★ Faster encryption and decryption
- ★ Lower power consumption
- ★ Better performance on low-cost processors
- ★ High security with less computational

Applications:

- ★ Smart home devices
- ★ Smart meters
- ★ Wearable health devices
- ★ Industrial IoT systems
- ★ Connected vehicles
- ★ Wireless sensor networks

Secure messaging process:

1. Two IoT devices generate ECC public and private keys
2. They exchange public key securely
3. Both devices compute the same shared secret using elliptic curve Diffie-Hellman
4. The shared secret is used to generate symmetric encryption key
5. Message encrypted and transmitted securely
6. The receiver decrypts the message using same shared key.

Advantage:

- * Strong security with small keys
- * faster computation
- * Lower battery consumption
- * Reduced bandwidth requirement

Conclusion:

elliptic -curve is one of the best choice for securing communication in IoT devices because it provide high security, efficient performance, low power consumption and minimal memory usage. It also enables secure messaging while meeting the resource of IoT environment.

SN